

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Fifth Semester of B. Pharm. Examination
University Theory Examination November/December 2018
PH319/PH334 Pharmaceutical Microbiology and Biotechnology

Date: 29/11/2018, Thursday **Time:** 10:00 a.m. to 01:00 p.m. **Maximum Marks:** 80

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat labeled sketches wherever necessary.
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SECTION - I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. The total magnification of a microscope is equal to _____.
[A] the magnification of the objective lens $\times$ the number of eye pieces
[B] the magnification of the objective $\times$ the number of ocular
[C] the total of all the objectives
[D] the price of the microscope
2. Sterilization is defines as _____.
[A] removal of microbial contamination
[B] complete destruction of all forms of life
[C] complete removal of vegetative cells
[D] removal of most microorganisms
3. Oligodynamic activity can be said to mean _____.
[A] a little bit works better than a lot
[B] very fast acting
[C] works well with other chemicals
[D] extremely toxic
4. Bacillus subtilis is an example of _____.
[A] gram positive bacteria
[B] gram negative bacteria
[C] virus
[D] viroid
5. Gram staining is an example for _____.
[A] simple staining
[B] differential staining
[C] negative staining
[D] spore staining

6. A facultative anaerobic bacteria is _____.
[A] only growing anaerobically
[B] only growing in the presence of oxygen
[C] ordinarily an anaerobe but can grow with oxygen
[D] ordinarily an aerobe but can grow in absence of oxygen
7. Rod shaped bacteria are known as _____.
[A] cocci
[B] comma forms
[C] bacilli
[D] pleomorphic forms
8. The action of alcohol during Gram-staining is to _____.
[A] allow the color
[B] add the color
[C] decolorize the cells
[D] intensify the color
9. Temperature required for pasteurization is _____.
[A] above 150 °C
[B] below 100 °C
[C] 130 °C
[D] 110 °C
10. Which of the following is ionizing radiation?
[A] UV rays
[B] IR
[C] γ -rays
[D] Visible light
11. The condition required for autoclave is _____.
[A] 121 °C temperature and 15 lbs. pressure for 15 minutes
[B] 100 °C temperature and 30 lbs. pressure for 30 minutes
[C] 150 °C temperature for 60 minutes
[D] 130 °C temperature for 120 minutes
12. Genetic constitution of the cell is _____.
[A] Phenotype
[B] Genotype
[C] Cryptotype
[D] Histotype
13. Yeast extract is an excellent source of _____.
[A] Vitamin A
[B] Vitamin B
[C] Proteins
[D] Carbohydrates
14. Rapid bacterial growth phase is known as _____ phase.
[A] Log
[B] Back
[C] Lag
[D] Cick

15. How much time a bacterium take for the complete duplication?
 [A] 30 minutes
 [B] 10 minutes
 [C] 20 minutes
 [D] 5 minutes
16. The best medium for the production of Penicillin is _____.
 [A] Nutrient Agar
 [B] Corn steep liquor
 [C] Sulfite waste liquor
 [D] Whey
17. The outstanding example of traditional microbial fermentation product is _____.
 [A] Vinegar
 [B] Penicillin
 [C] Citric acid
 [D] Tetracycline
18. In the industrial production of streptomycin, the secondary metabolite or byproducts is _____.
 [A] Vitamin B₁₂
 [B] Vitamin C
 [C] Vitamin B₆
 [D] Ethanol
19. To differentiate lactose and non-lactose fermenters, the medium used is _____.
 [A] Mac Conkey's medium
 [B] Stuart's medium
 [C] Sugar medium
 [D] Citrate medium
20. Plasmids are ideal vectors for gene cloning as _____.
 [A] They can be multiplied by culturing
 [B] They can be multiplied in the laboratory using enzymes
 [C] They can replicate freely outside the bacterial cell
 [D] They are self-replicating within the bacterial cell

SECTION – II

- Q 2** Attempt any **TWO** of the following
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|----------|---|-----------|
| A | Draw a sketch of an autoclave. Explain the principle of working for an autoclave. Enlist the various applications of an autoclave. | 05 |
| B | Describe the remarkable contributions of great scientists (at least five) to the field of pharmaceutical microbiology. | 05 |
| C | Explain the two cardinal objectives for classifications of microorganisms. Enlist the three most prominent genetic methods employed for the methodical arrangement of microbes based upon various taxonomic groups. Explain any one method in detail. | 05 |

- Q 3** Attempt any **TWO** of the following
- A** Draw and explain in detail the pattern of normal growth curve (or growth cycle) of microorganisms incubated for growth in due course of time. **05**
- B** Write a note on mechanism of action of phenol, halogens and heavy metals for their disinfectant property. **05**
- C** Write note on recombinant DNA technology. **05**

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Enlist the recognized and well-established methods for the isolation of bacteria. Describe the use of the McConkey's medium as a selective and diagnostic media for isolation of Enterobacteriaceae from faeces, urine, foods and water. **05**
- B** Explain the cardinal factors invariably and predominantly affecting the efficacy of disinfectants. **05**
- C** Krihesha Pharma LLC is involved in manufacturing of 0.9 %W/V NaCl Large volume parenteral, Chloramphenicol Eye Drops, Surgical sutures, Neomycin sulphate powder and Testosterone IM injection. Suggest the appropriate method/s of sterilization for the mentioned pharmaceutical products. **05**
- D** Enlist the well-recognized and universally accepted methods for sterility testing of pharmaceutical products. Explain, in-detail, any one method of sterility test. **05**
- E** Define microbial biotransformation. Explain the types of microbial transformation for steroids. **05**
- F** Describe the microbial assay of vitamin B₁₂. **05**
- Q 5** Attempt any **FOUR** of the following
- A** Explain the mechanisms of action, advantages and disadvantages of alcohols as effective bactericidal and fungicidal agents. Enlist the names of two most frequently employed alcohols. **05**
- B** Explain the modes/mechanisms of action of antimicrobial agents to inhibit or kill the microorganisms. **05**
- C** Write note on microbial limit testing of pharmaceutical dosage forms. **05**
- D** Describe one level/dimensional microbial assay of streptomycin. **05**
- E** Write note on production of penicillin using fermentation process. **05**
- F** Enumerate the advantages of immobilized enzymes. Explain the different methods of enzyme immobilization. **05**